



Instruction sheet 739 5821

Training panel lighting NG



1 Description

The training panel consists of a modern dash panel insert with electronic immobilizer, complete steering wheel, the central module comfort system and the electronic central electronics module. The lighting system and a wiper motor complete the system. Fundamentals of automotive electronics and modern databus systems are presented clearly and logically. Great importance is attached to this concept by using original vehicle components. The error detection, fault analysis and correction are on the focus. For data acquisition and analysis of the data messages the CASSY system is used. The data messages are recorded and interpreted on an oscilloscope basis. The system is self-diagnostic capability, an appropriate diagnostic tester ([737 9803](#)) can be played through all the diagnostic possibilities.

1.1 Technical data

Supply voltage 13.8 V=
Maximum power consumption 40 A

2 Additional equipment

2.1 Accessories

No.	Material	Cat. number
1	Sensor-CASSY 2	524 013
1	CAN-Databus-Multi-Adapter	773 961
1	LIN-Bus PC Interface USB	739 5881

3 Recycling and disposal



Separate collection of electrical and electronic equipment

In many countries around the world, electronic devices are separated and recycled due to the valuable raw materials they contain, and are not disposed of with household waste.

Used batteries and accumulators that are not enclosed within the device, as well as lamps that can be removed from the device without damage, must be separated from the device before disposal. Personal data must also be deleted from the devices to be disposed of.

To dispose of these devices within Germany, the used devices and removed electronic components should be sent or taken to the following address:
LD DIDACTIC GmbH – Service
Gleueler Straße 72
D-50226 Frechen

Within the EU, the WEEE Directive 2012/19/EU applies. For disposal inquiries, please contact your dealer.

In all other countries, please check the relevant national regulations.

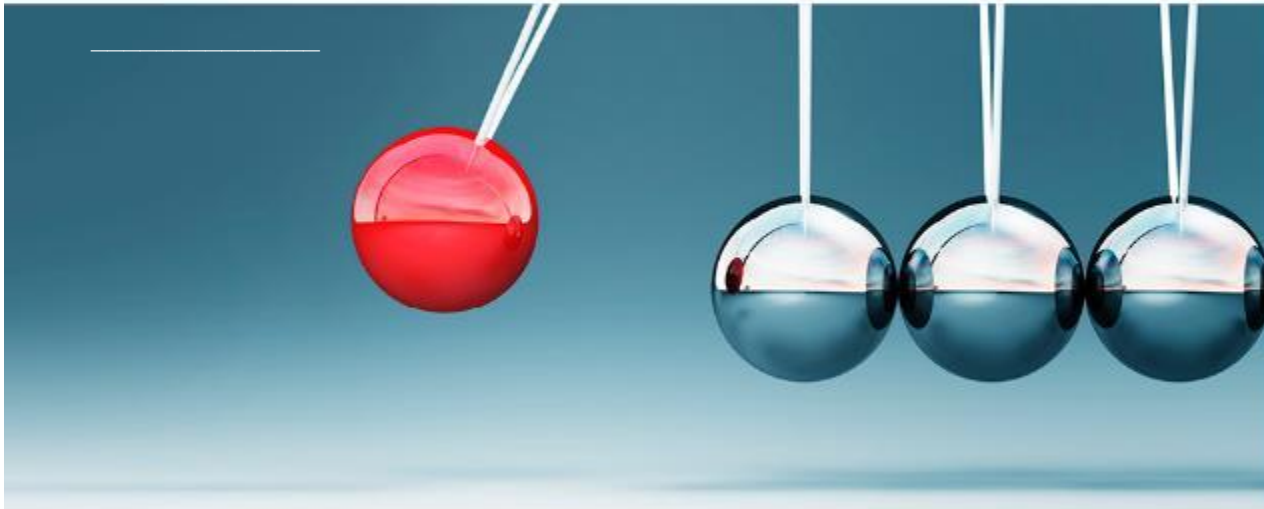
4 Appendix



LD DIDACTIC

NETWORKING SYSTEMS IN MODERN VEHICLES

739 5821: Training panel lighting



- Description and setup
- Tasks
- Worksheets

VERSION 2016.1

In cooperation with Audi AG, Germany



 **Description and Construction**

 **Exercises**

 **Participant Documents**

Description and Construction

This training panel is a section of the electronic systems network for the Audi A4, model year 2007.

This system consists of a dash board with immobilizer in high-line level, steering wheel electronics, the central comfort system module and the Central Electronics unit. The lighting system is represented using commercially available lamps. The window washer motor is hidden behind the wheel bolt cover.

The training panel help you to understand the logic of how the electronic systems are networked.

To start up the training panel, connect it to the 12VDC power supply.

The current required is max. 40 A. Thus an automotive battery is the ideal voltage source.



Power supply: 12 V=
Ground: black socket
Plus: red socket



Diagnostics plug -T16 for the diagnosis interface 7379803

Since the training panel only represents a section of the electronic systems network for the Audi A4, the diagnosis will recognize the following faults when performing the „automatic test sequence“. This is because the control modules on the training panel expect a message from all control modules which are necessary for operating the vehicle at „Ignition on“

Address word 09: Electronic Central Electrics „3 faults detected“ Code 01

- 01331 Door control module driver side -J386 no communication
- 01320 Control Module for Climatronic-J255 no communication
- 00474 Control unit immobilizer – no communication

Address word 16: Steering wheel Wheel Electronics „2 faults detected“ Code 11

- 01331 Door control module driver side -J386 no communication
- 01332 Door control module passenger side -J387 no communication

Address word 17: Dashboard insert „4 faults detected“ Code 11

- 01314 Engine Control Module no communication
- 01331 Door control module driver side -J386 no communication
- 01332 Door control module passenger side -J387 no communication
- 01128 Coil for immobilizer



Diagnostics plug -T16 for VCDS

Address word 25: Immobilizer

- see Address word 17: Switchboard insert

Address word 46: Comfort System Central Module „5 faults detected“ Code 0

- 01320 Control Module for Climatronic-J255 no communication
- 01331 Door control module driver side -J386 no communication
- 01332 Door control module passenger side -J387 no communication
- 01366 Opening from Crash-Signal interruption
- 01134 Alarm horn H12 – no communication

Note

Further unknown faults may occur!



CAN databus interface



Red:
Black:
High

CAN Low
CAN

On this port you can use a CAN databus interface and the adapter cable to send the CAN-bus signals to your computer. Software visualizes the signals.



Lighting system I

Consists of two headlights, high and low beam, the backup lights and the turn signals, two fog lamps, the third brake light and the licence plate light.

Wiper motor - V,
Hidden behind the wheel bolt cover.

Switch for reverse gear - F4

Ignition lock - D

Switch for the warning flashers - E229



Lighting system II

Sockets for front lights:

M1	Left side light bulb
M5	Front left turn signal bulb
M29	Left dipped beam bulb
M30	Left main beam bulb

Sockets for rear lights:

M17	Right reversing light bulb
M8	Rear right turn signal bulb
L47	Rear right fog light bulb
M2	Right tail light bulb
M10	Right brake light bulb

Note:

Please be aware that due to construction measures the left front light unit is located on the right side of the board and vice versa!

Light switch -
E1

Switch for
fog lamps and
tail light - E23

The Steering wheel Column Control Module - J527,
With remote control buttons

Brake light switch - F





Dash panel insert - J285,
with the control module for the immobilizer and the
gateway as an interface for the drive bus and the
display bus.

Central control unit for convenience
System - J393,

Control unit for vehicle electrical
System - J519,

Fuses



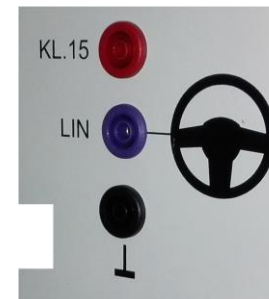
Control unit Automatic Trailer Detection - J471
This ECU controls the pins of the 13 pole trailer socket depending on the light switch and switches off the rear fog lights of the car

Trailer socket
A 13 pole trailer socket to which i. e. a bicycle rack can be connected to



The rain light sensor with measurement Sockets for the LIN databus signal

LIN databus signal of the steering wheel



Additional connection in use with the original vehicle door (see next page)



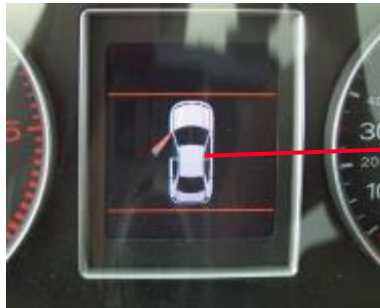
Additional signal for the door(s) 7395835 and/or 7395836
(Schematics path B207)



Connection door(s) B207

Connection CAN-H (black) and CAN-L (red)

Connection Ground
Connection T30 (Plus)



Display „door open“



Door lock

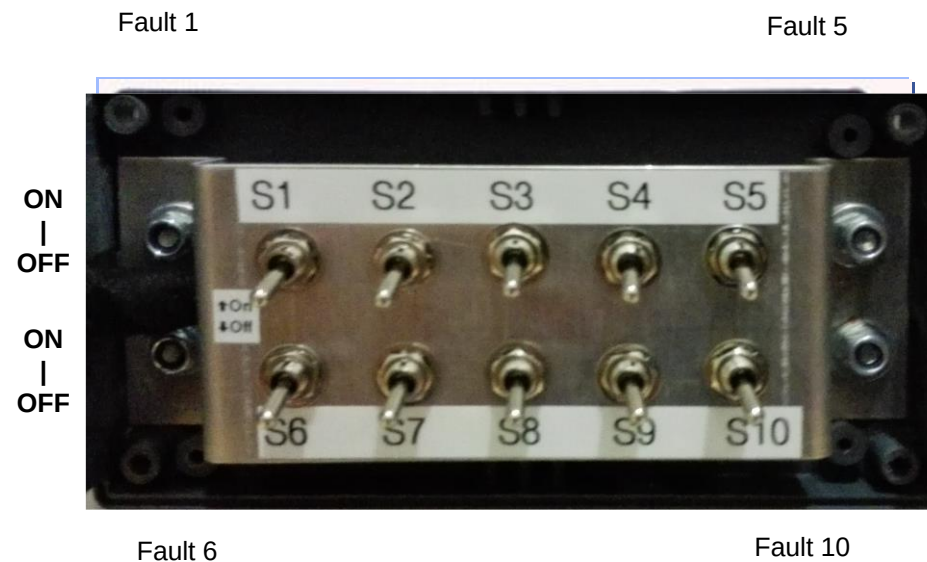


Window and mirror adjustment keys



The switch box on the rear side

for fault simulation:



Fault switch 1

Description: Rain light sensor does not switch off the light in position „AUTO“ (T30 interruption)
 Diagnostic note: (Adress word 09)

Fault switch 2

Description : Side direction indicator left (T30 interruption)

Fault switch 3

Description : Rear fog light right (T30 interruption)
 Diagnostic note : (Adress word 09)

Fault switch 4

Description : Interruption of pin 3 from ILM to front wiper motor, no operation of the motor in stage 1
 Diagnostic note : No fault entry neither in dash board nor in diagnostic tester!

Fault switch 5

Description : High beam right: T30 interruption
Diagnostic note : M 32 Interruption/ short to plus (Adress word 09)

Fault switch 6

Description : Trailer back up light: T30 interruption
Diagnostic note : No fault entry neither in dash board nor in diagnostic tester!

Fault switch 7

Description : Fog lights and rear foglights are not functioning in position 3 of the rotary light switch
Diagnostic note : (Adress word 09)

Fault switch 8

Description : CAN High Short to plus (T. 30)

Fault switch 9

Description : CAN Low Short to ground (T. 31)

Fault switch 10

Description : CAN Low and CAN High shortened

Diagnostic note : Konzern – Databus Comfort in single wire operation (Adress word 09)

NOTE: the faults #8 and #9 must never be activated simultaneously!

Excercises



CAN databus

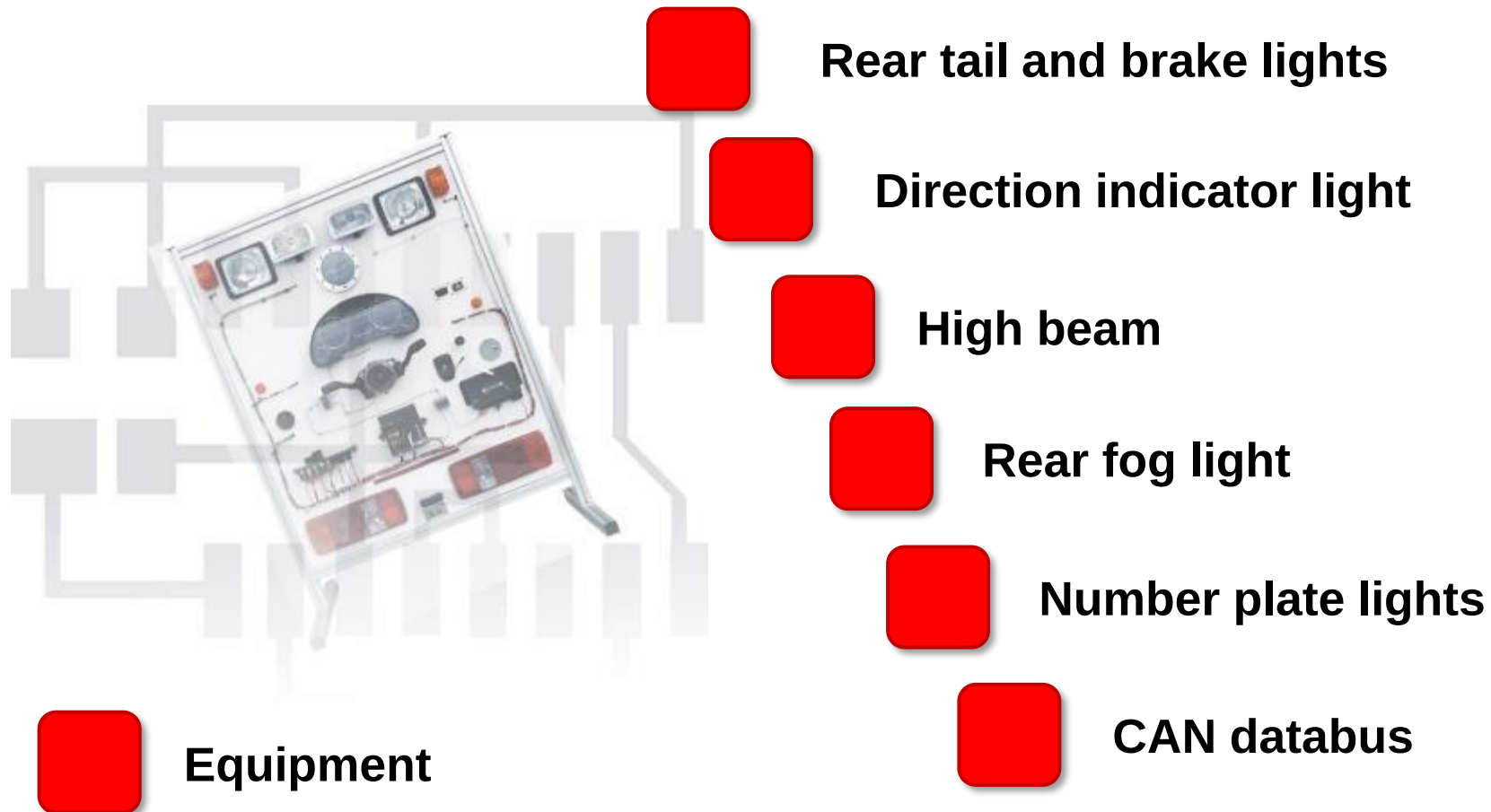


LIN databus



Diagnosis

CAN databus excercises



NETWORK IN THE A4 ON THE COMFORT DATABUS

■ Messages used for tail light control

Sender		Identi- fication	Data length	Send interval	J519	J285	
Electron. Cent. Elec.	- J519	531h ¹	2 bytes	50ms	S	E	
Electron. Cent. Elec.	- J519	631h ²	4 bytes	1s	S	E	

¹ = also: sent as soon as there is a change
² = Sent only when there is a fault message
 S= Sender
 E= Receiver

Notes for fault messages in estate car:
 •Lamps are checked after ignition „on“ and during operation
 •A healing process is possible essentially analogous to the above point

TAIL/BRAKE LIGHT

■ Ignition on - Parking light off

Light
switch



Light Off

Electron.
Central El.



531h

00 00 h

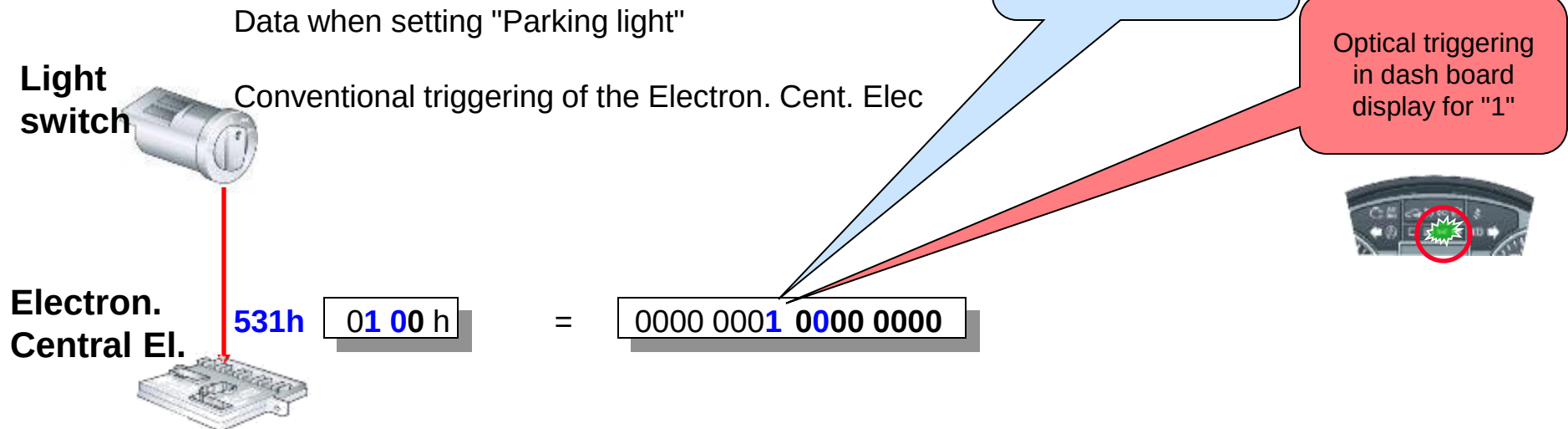
=

0000 0000 0000 0000

Parking light off

TAIL/BRAKE LIGHT

■ Parking light is set



NETWORK IN A4 ON COMFORT DATABUS

■ Messages used for turn signal control

Sender	Identi- fication	Data length	Send interval	J527	J519	J285
Steering wheel Column - J527	2C1h ¹	3 bytes	100ms	S	E	E
Electron. Cent. Elec. - J519	531h ¹	2 bytes	50ms		S	E
Electron. Cent. Elec. - J519	631h ²	4 bytes	1s		S	E

¹ = also: sent as soon as there is a change

² = Sent only when there is a fault message

S= Sender

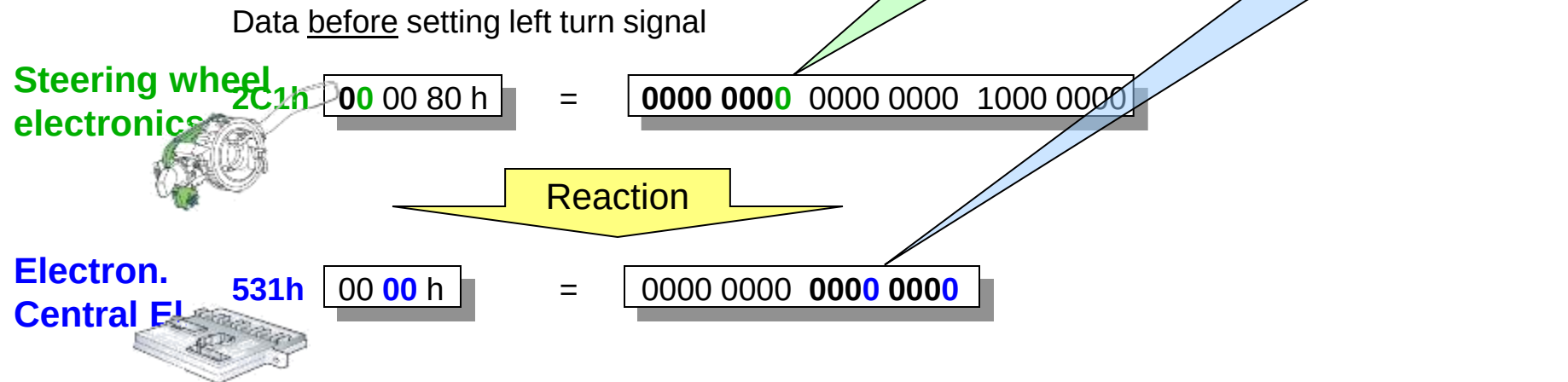
E= Receiver

Notes for fault messages in display:

- Lamps are checked after ignition „on“ and during operation
- A healing process is possible essentially analogous to the above point

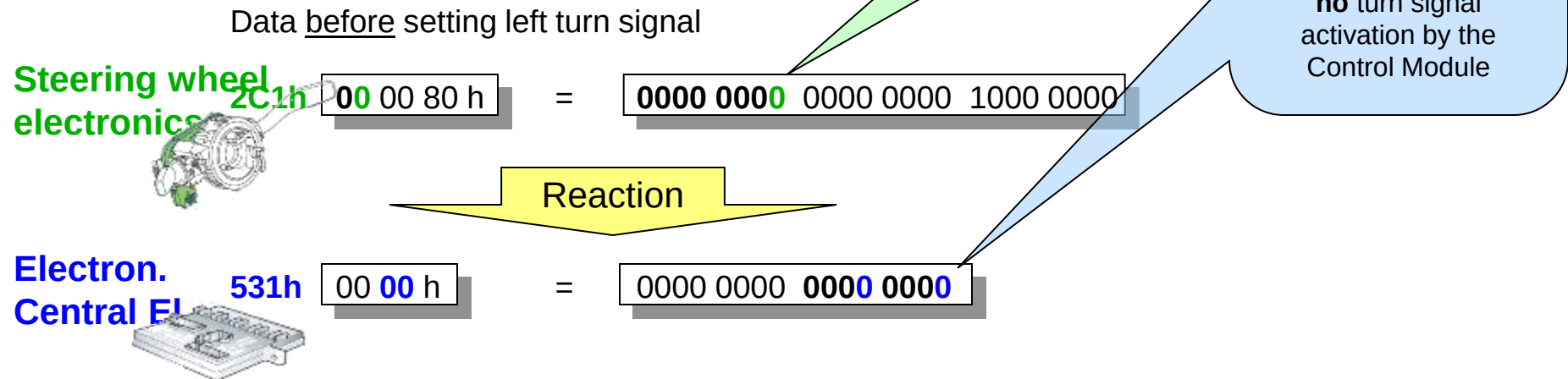
TURN SIGNAL CONTROL

■ No turn signal activity



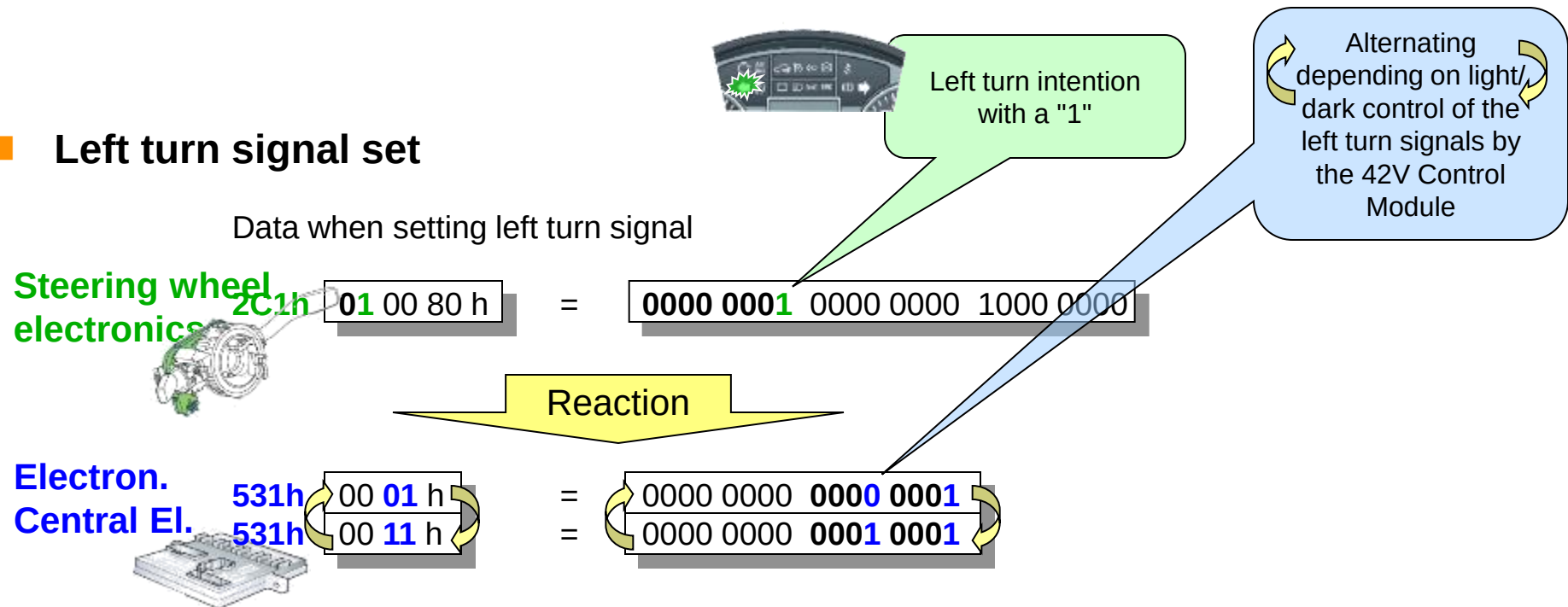
TURN SIGNAL CONTROL

■ No turn signal activity



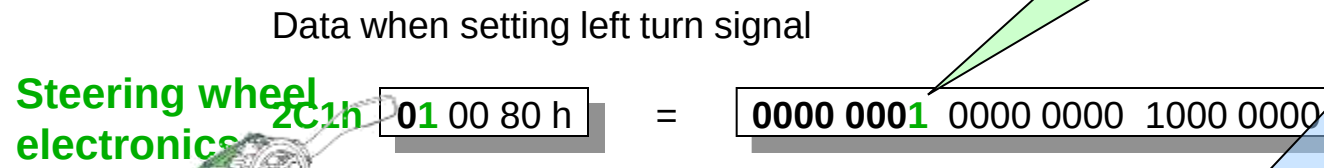
TURN SIGNAL CONTROL

■ Left turn signal set

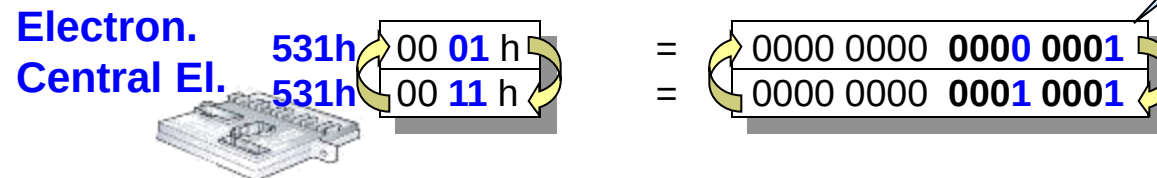


TURN SIGNAL CONTROL

■ Left turn signal set



Reaction



Left turn intention with a "1"

Response: Left turn signal active

Optical (left) and acoustic triggering in dash board for "1"



TURN SIGNAL CONTROL



■ Interrupted left side turn signal (fault 2)

Data during flashing

Steering wheel
electronics



2C1h

01 00 80 h

=

0000 0001 0000 0000 1000 0000

Reaction

Electron.
Central El.



531h

00 01 h

=

0000 0000 0000 0001

531h

00 11 h

=

0000 0000 0001 0001

631h

04 00 00 00 h

=

0000 0100 0000 0000 0000 0000 0000 0000

Alternating
depending on light
dark control of the
left turn signals by
the control Module

TURN SIGNAL CONTROL

■ Interrupted left side turn signal (fault 2)



Data during flashing

Steering wheel
electronics



2C1h

01 00 80 h

=

0000 0001 0000 0000 1000 0000

Reaction

Electron.
Central El.



531h

00 01 h

=

0000 0000 0000 0001

531h

00 11 h

=

0000 0000 0001 0001

631h

04 00 00 00 h

=

0000 0100 0000 0000 0000 0000 0000 0000

Alternating
depending on light
dark control of the
left turn signals by
the control Module

Fault messages
as long as no OK
recognized

Result

Fault message in
dash board
display for "1"



TURN SIGNAL CONTROL



■ Left front turn signal - Fault cleared

Data during flashing

Steering wheel
electronics



2C1h 01 00 80 h

= 0000 0001 0000 0000 1000 0000

Reaction

Electron.
Central El.



531h 00 11 h
531h 00 01 h

= 0000 0000 0001 0001
= 0000 0000 0000 0001

631h 04 00 00 00 h
631h 00 00 00 00 h

= 0000 0100 0000 0000 0000 0000 0000 0000
= 0000 0000 0000 0000 0000 0000 0000 0000

Alternating
depending on light
dark control of the
left turn signals by
the control Module

Left turn signal active

TURN SIGNAL CONTROL

■ Left front turn signal - Fault cleared



Data during flashing

Steering wheel electronics



2C1h

01 00 80 h

Reaction

Electron. Central El.



531h

00 11 h

531h

00 01 h

631h

04 00 00 00 h

631h

00 00 00 00 h

= 0000 0001 0000 0000 1000 0000

= 0000 0000 0001 0001

= 0000 0000 0000 0001

= 0000 0100 0000 0000 0000 0000 0000 0000

= 0000 0000 0000 0000 0000 0000 0000 0000

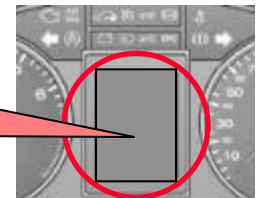
Alternating depending on light/dark control of the left turn signals by the control Module

Left turn signal active

approx. 3 OK messages after recognized fault clearing plus left flashing

Result

No Fault message in dash board display when „0“



Attention!

This experiment requires the 12 V power supply 738 026/027!

Safety notes

Never look directly into the high beam!

I've read the safety notes

E= Receiver

- A healing process is possible essentially analogous to the above point

HIGH BEAM CONTROL

- Ignition, driving light is turned on - high beam is off

Data before setting „High beam“

Steering wheel
electronics



2C1h

00 00 80 h

=

0000 0000 0000 0000 1000 0000

Reaction

Electron.
Central El.



531h

03 00 h

=

0000 0011 0000 0000

HIGH BEAM CONTROL

■ Ignition, driving light is turned on - high beam is off

Data before setting „High beam“



2C1h

00 00 80 h

=

0000 0000 0000 0000 1000 0000

Reaction

Electron.
Central El.



531h

03 00 h

=

0000 0011 0000 0000

Turn signal switch in
neutral position

High beam **off** on
Control Module

HIGH BEAM CONTROL

■ High beam is set

Steering wheel
electronics



2C1h

08 00 80 h

=

0000 1000 0000 0000 1000 0000

Electron.
Central El.



531h

07 00 h

=

0000 0111 0000 0000

Reaction

High beam On
by means of "1"

Response from
High beam triggering

Optical triggering
in dash board
display for "1"



HIGH BEAM CONTROL



■ Interruption in right high beam lamp (Fault 5)

Data during High beam On

Steering wheel
electronics



2C1h

08 00 80 h

=

0000 1000 0000 0000 1000 0000

Reaction

Electron.
Central El.



531h

07 00 h

=

0000 0111 0000 0000

631h

08 00 00 00 h

=

0000 1000 0000 0000 0000 0000 0000 0000

Fault messages
as long as no OK
recognized

Result

Fault message in
dash board
display for "1"



HIGH BEAM CONTROL



■ Left high beam lamp - Fault cleared

Data during High beam On

Steering wheel
electronics



2C1h

08 00 80 h

=

0000 1000 0000 0000 1000 0000

Reaction

Electron.
Central El.



531h

07 00 h

=

0000 0111 0000 0000

631h

08 00 00 00 h

=

0000 1000 0000 0000 0000 0000 0000 0000

631h

00 00 00 00 h

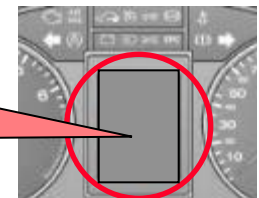
=

0000 0000 0000 0000 0000 0000 0000 0000

High beam Off - On !!
approx. 3 OK messages
after recognized fault
clearing

Result

No Fault message
in dash board
display when „0“



NETWORK IN A4 ON COMFORT DATABUS

■ Messages used for rear fog light control

Sender		Identi- fication	Data length	Send interval	J519	J285	
Electron. Cent. Elec.	- J519	531h ¹	2 bytes	50ms	S	E	
Electron. Cent. Elec.	- J519	631h ²	4 bytes	1s	S	E	

¹ = also: sent as soon as there is a change

² = Sent only when there is a fault message

S= Sender

E= Receiver

Notes for fault messages in the display:

- Lamps are checked after ignition „on“ and during operation
- A healing process is possible essentially analogous to the above point

REAR FOG LIGHT CONTROL



■ Ignition and main lights on – rear fog lights off

Data before setting „Rear fog light“

Light
switch



Rear fog light Off

Electron.
Central El.



531h

00 00 h

=

0000 0000 0000 0000

Rear fog light **off**

REAR FOG LIGHT CONTROL

■ Rear fog light is set



Data when setting „Rear fog light“

Light
switch



Conventional triggering of the Electron. Cent. Elec.

Electron.
Central El.



531h

01 00 h

=

0000 0001 0000 0000

Response from
Rear fog light control

REAR FOG LIGHT CONTROL



■ Defective rear fog light lamp-message, fault 3

Data while rear fog lights are On

Light switch



Conventional triggering of the Electron. Cent. Elec.

Electron. Central El.



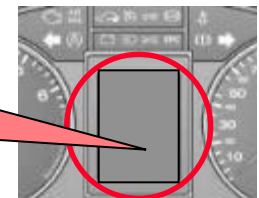
531h 01 00 h = 0000 0001 0000 0000

631h 00 00 10 00 h = 0000 0000 0000 0000 0001 0000 0000 0000

Fault message

Result

Fault message in dash board display when receiving „1“



REAR FOG LIGHT CONTROL



■ Rear fog light lamp - Fault cleared

Data while rear fog lights are On

Light switch



Conventional triggering of the Electron. Cent. Elec.

Electron.
Central El.



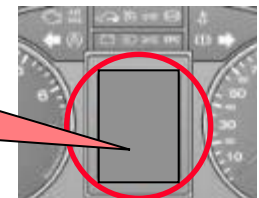
531h 01 00 h = 0000 0001 0000 0000

631h 00 00 10 00 h = 0000 0000 0000 0000 0001 0000 0000 0000
631h 00 00 00 00 h = 0000 0000 0000 0000 0000 0000 0000 0000

approx. 3 OK
messages after
recognized fault
clearing

Result

No Fault message
in dash board
display when „0“



NETWORK IN A4 ON COMFORT DATABUS

■ Messages used for licence plate control

Sender		Identi- fication	Data length	Send interval	J519	J285	
Electron. Cent. Elec.	- J519	531h ¹	2 bytes	50ms	S	E	
Electron. Cent. Elec.	- J519	631h ²	4 bytes	1s	S	E	

¹ = also: sent as soon as there is a change

² = Sent only when there is a fault message

S= Sender

E= Receiver

Notes for fault messages in display :

- Lamps are checked after ignition „on“ and during operation
- A healing process is possible essentially analogous to the above point

LICENCE PLATE LAMP CONTROL

■ Ignition on - parking light off

Data before setting „Parking light“

Light
switch



Light Off

Electron.
Central El.

531h

00 00 h



=

0000 0000 0000 0000

Parking light **off**

LICENCE PLATE LAMP CONTROL

■ Parking light is set



Data when setting "Parking light"

Light
switch



Conventional triggering of the Electron. Cent. Elec.

Electron.
Central El.



531h

01 00 h

=

0000 0001 0000 0000

Response from
parking light
triggering

Optical triggering
in dash board
display for "1"



LICENCE PLATE LAMP CONTROL



Defective licence plate lamp - message

Data while parking light is On

Light switch



Conventional triggering of the Electron. Cent. Elec.

Electron. Central El.



531h

01 00 h

=

0000 0001 0000 0000

631h

00 00 40 00 h

=

0000 0000 0000 0000 0100 0000 0000 0000

By means of PCANView software transmit this message every 200 ms

Fault messages since lamp stage is switched off

Result

Fault message in dash board display for "1"



LICENCE PLATE LAMP CONTROL



■ Licence plate lamp - Fault cleared

Data while parking light is On

Light switch



Conventional triggering of the Electron. Cent. Elec.

Electron. Central El.



531h 01 00 h = 0000 0001 0000 0000

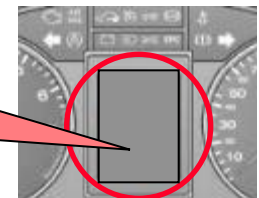
631h 00 00 10 00 h = 0000 0000 0000 0000 0100 0000 0000 0000
631h 00 00 00 00 h = 0000 0000 0000 0000 0000 0000 0000 0000

By means of PCANView software now transmit this message every 200 ms

approx. 3 OK messages after recognized fault clearing

Result

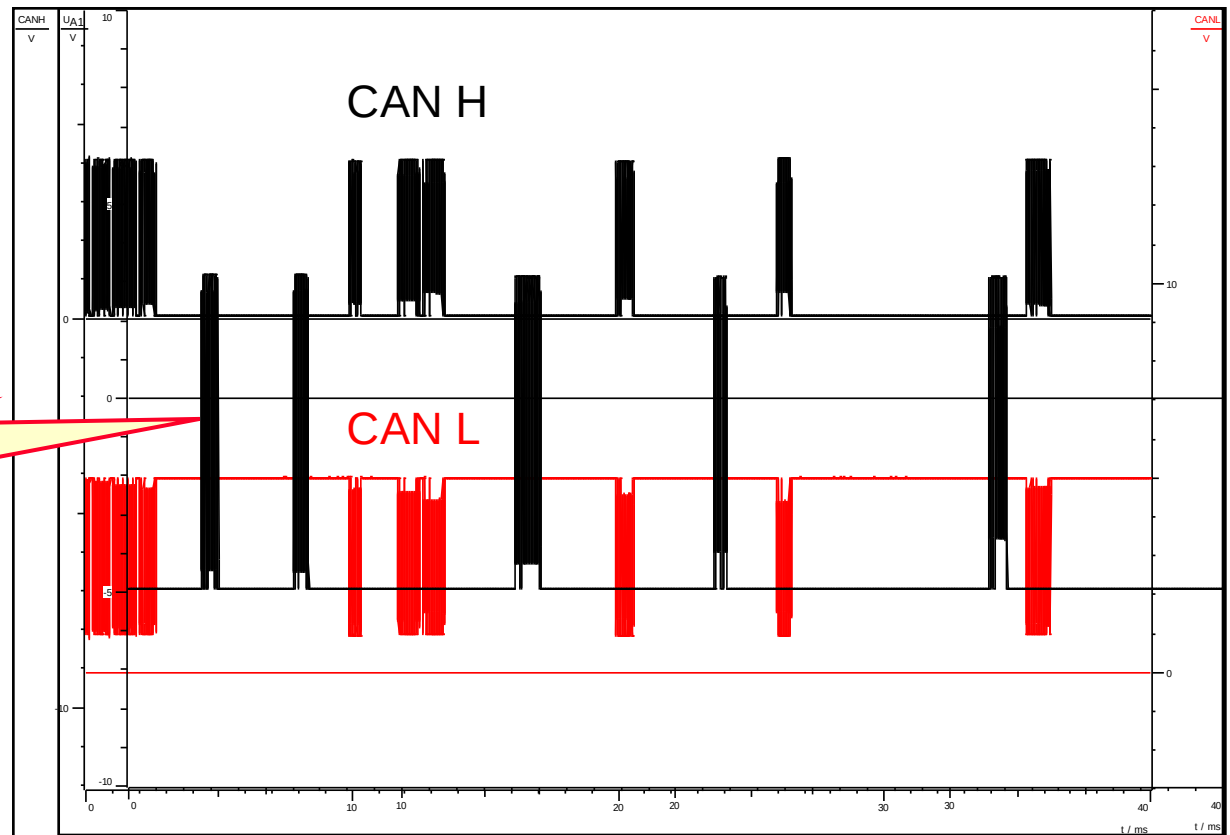
No Fault message in dash board display when „0“



THE CAN DATABUS

- Signals of CAN_H and CAN_L

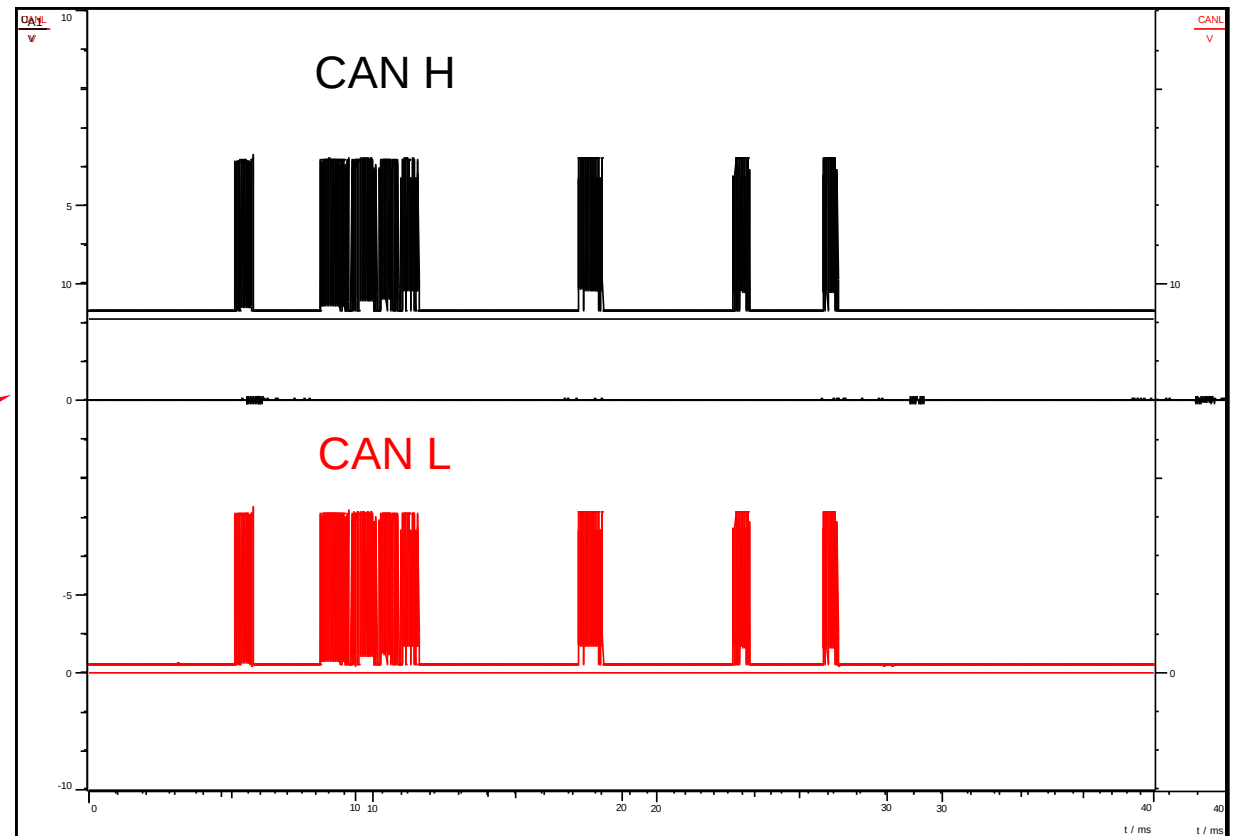
Measurement between
CANH and CANL



THE CAN DATABUS

- Signals in case of fault #10: short circuit

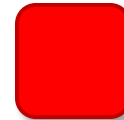
Measurement between
CANH and CANL



LIN DATABUS EXERCISES



Wiper



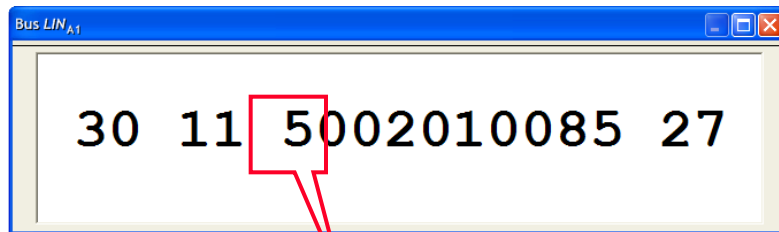
Steering wheel remote



Equipment

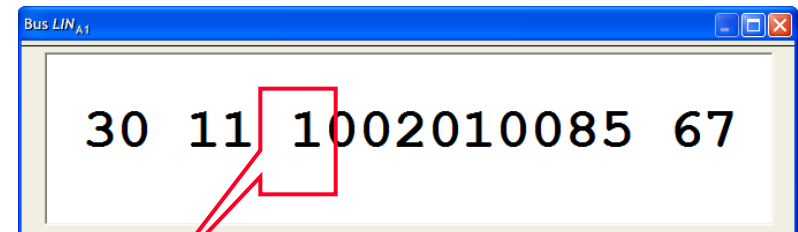
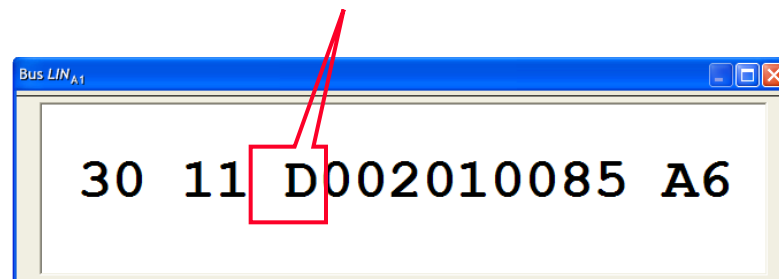
RAIN-LIGHT SENSOR

■ LIN databus messages

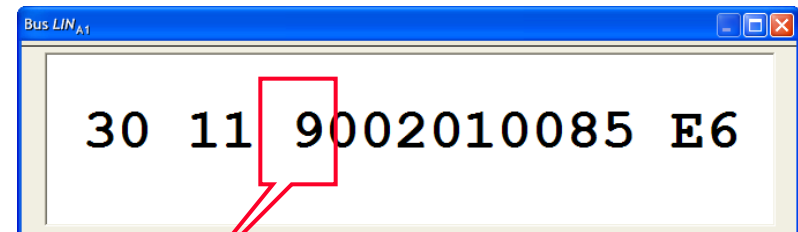


Interval 2

Interval 4

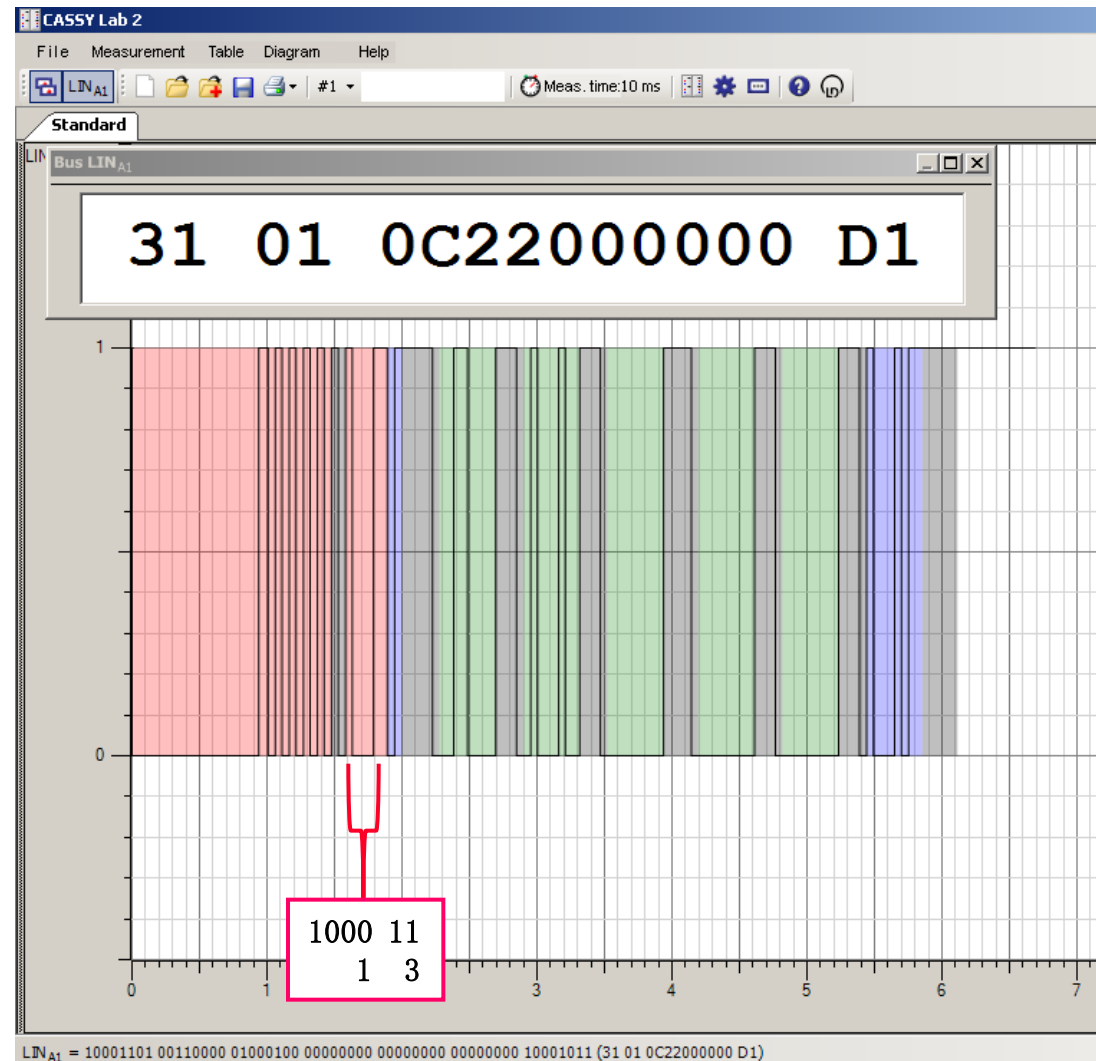


Interval 1



Interval 3

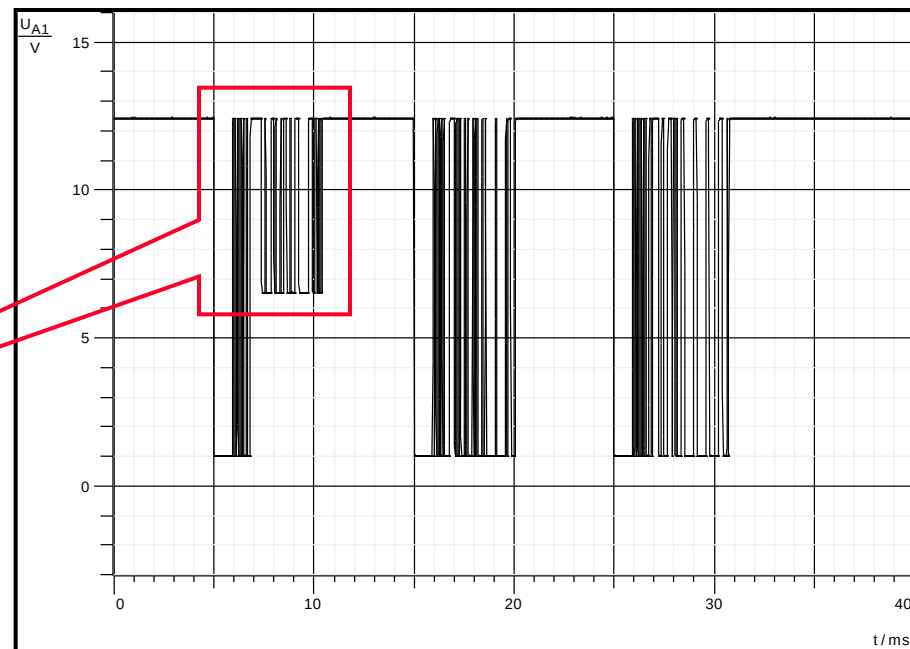
- Select message 31 and take a measurement and make an interpretation of the identifier!



LIN databus fault

- Switch on fault #1
- Take an oscillogram of the LIN signal between central electric control unit J519 and wiper motor

Reduced voltage level
due to contact resistance

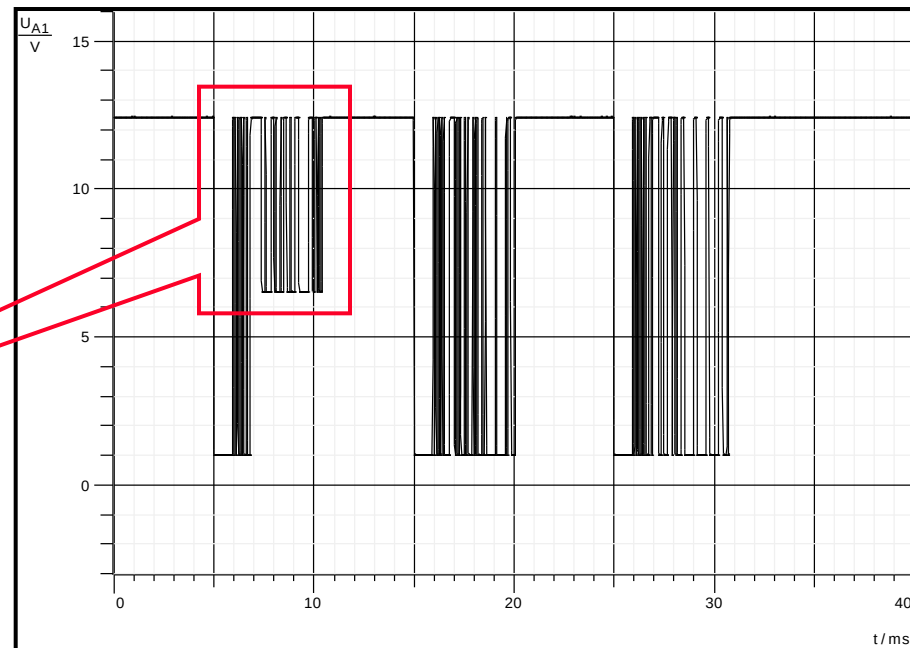


STEERING WHEEL REMOTE SIGNALS

LIN databus fault

- Switch on fault #10
- Take an oscillogram of the LIN signal between central electric control unit J519 and wiper motor

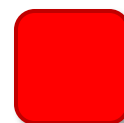
Reduced voltage level
due to contact resistance



DIAGNOSIS EXERCISES



CAN databus



LIN databus



Lighting



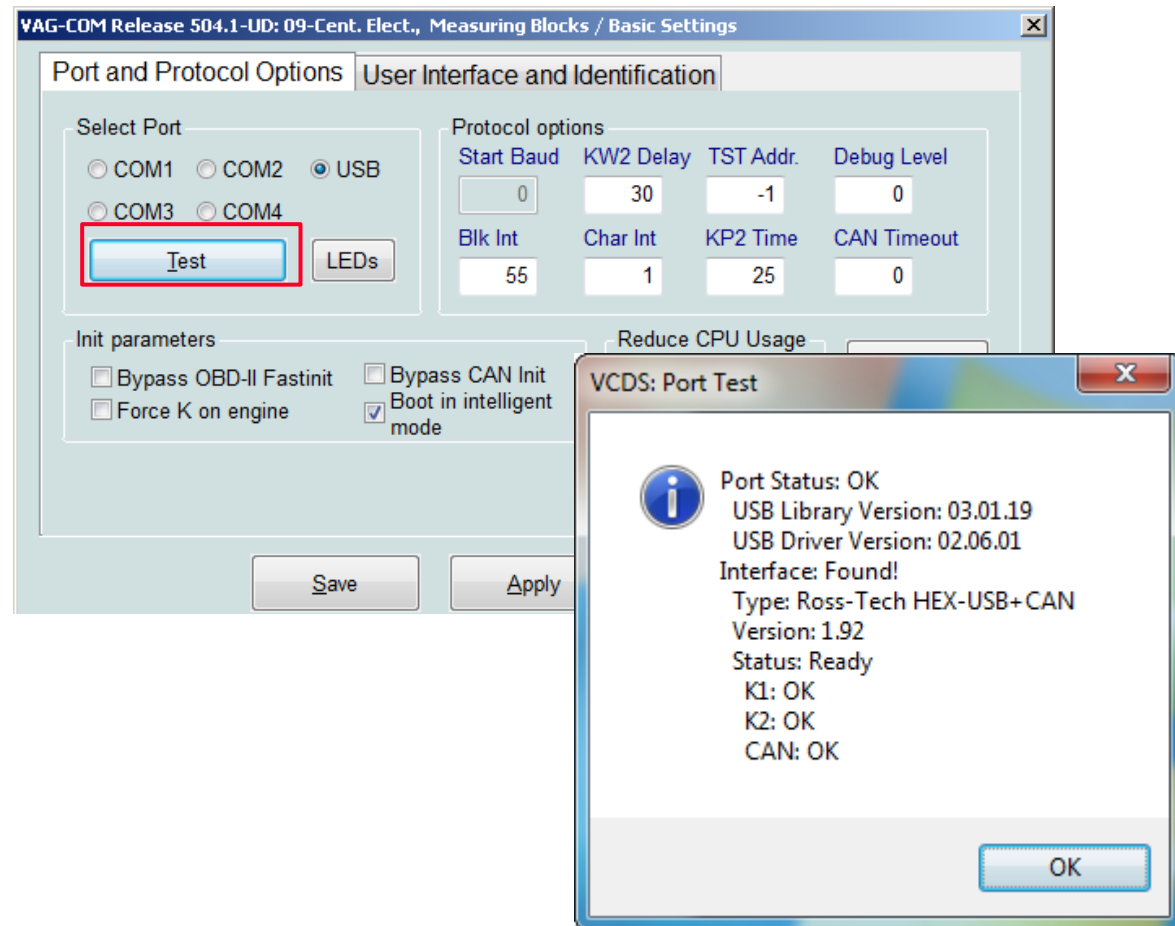
Equipment



Notes on coding

INTRODUCTION

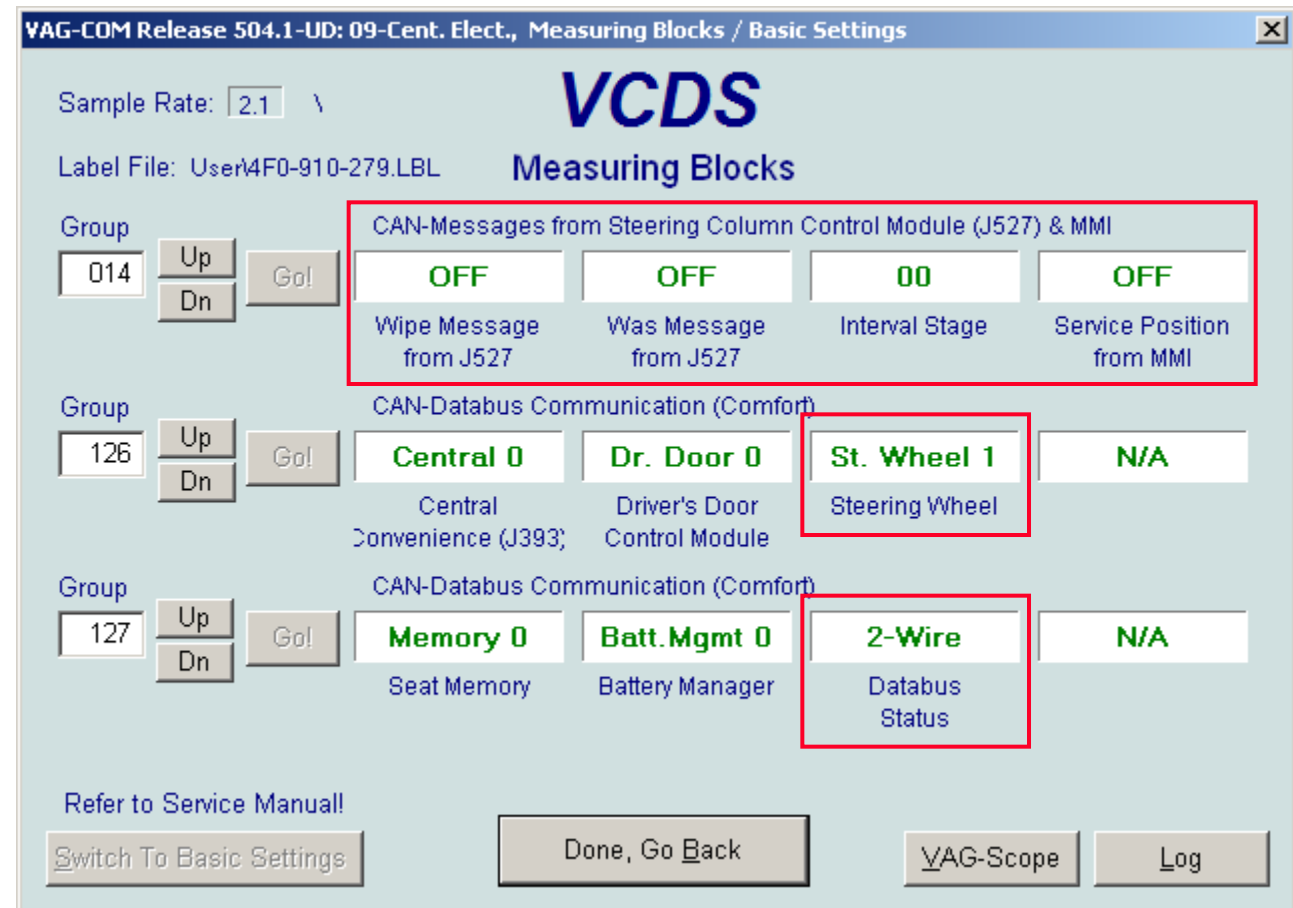
- **Run VCDS software!**
- Make sure the ignition is turned on and the interface plugged into the vehicle (see p. 3), then click **[Test]** to see if VCDS can find the interface.



DIAGNOSIS WITH VCDS

„Reading measuring blocks“

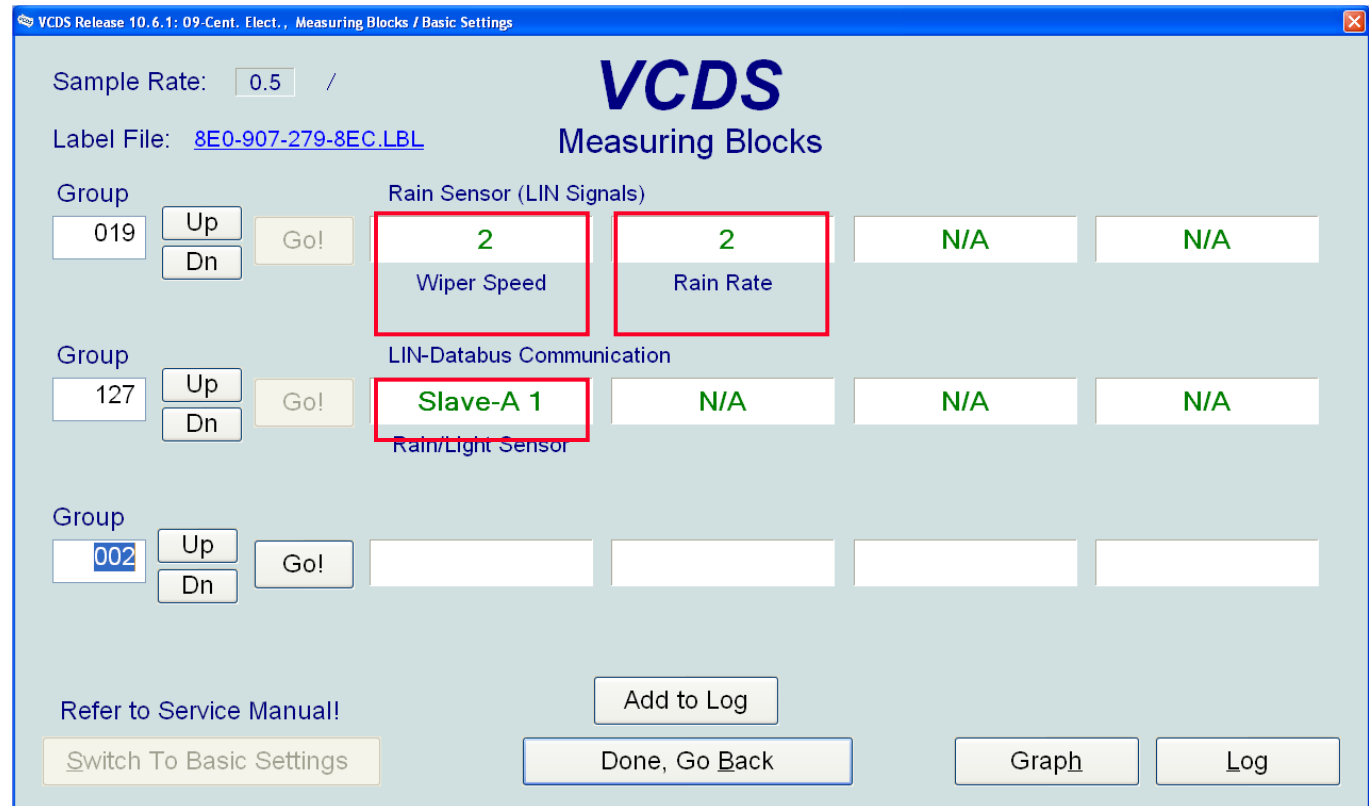
- Read group 14, values 126 and 127
- Interrupt a CAN bus wire and observe group 127, value



DIAGNOSIS WITH VCDS

„Measuring Blocks“ (LIN databus)

- Read blocks 19 and 127
- Wipe repeatedly with your hand over the rain light sensor cover
- Observe the values shown in the software!



DIAGNOSIS WITH VCDS

„Read fault codes – Auto Scan“

- Select [Auto Scan]
- Select „Audi A4 4F“
- Run [Start]

VAG-COM Release 504.1-UD: 09-Cent. Elect., Measuring Blocks / Basic Settings

Note: Chassis Types can be modified by editing the [MyAutoScan.TXT file](#)

Select Chassis Type:

8E - Audi A4/S4/RS4 B6/t ▼

☒ UDS aggressive mode

☒ Display freeze frame

Start Stop

Gateway Installation List

☐ Auto Refresh

Clear all DTCs

Results

Copy Print

Save Clear

Close

Chassis Type: 4F (8E - Audi A4/S4/RS4 B6/B7 (2001 > 2008))

Scan: 01 02 03 08 09 0F 11 15 16 17 18 25 36 37 45 46 55 56 57 65 67 69 75 76 77

VIN: WAUZZZ4F36N111022 Mileage: 640km-397miles

Address 01: Engine Labels: 06F-907-115-AXX.lbl

Part No SW: 4F2 910 115 B HW: 4F2 907 115

Component: 2.0l R4/4V TFSI 01 0010

Revision: -H11- Serial number:

Coding: 0105000318030018

Shop #: WSC 02325 768 00200

VCID: 1F46FC54CEAED0C7E0-804A

20 Faults Found:

012361 - Control Circuit for Starter Relay 2

P3049 - 004 - Open Circuit

Freeze Frame:

Fault Status: 01100100

Fault Priority: 0

Fault Frequency: 1

Reset counter: 255

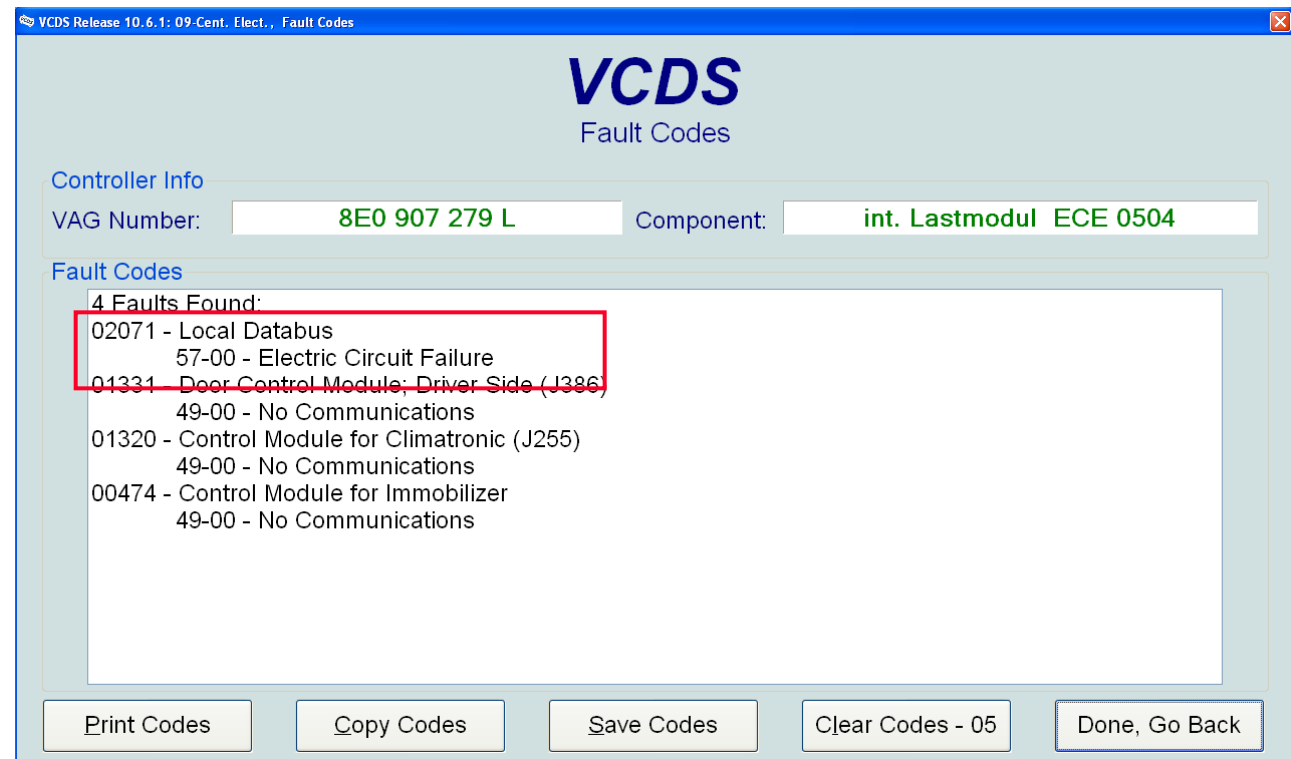
Mileage: 0 km

Time Indication: 0

DIAGNOSIS WITH VCDS

„Read fault codes“
(LIN databus)

- Interrupt the LIN connection of the wiper motor!
- Read fault code (Addr. 02)



DIAGNOSIS WITH VCDS

„Adaptation“

- Set the daytime running light dimming value to 92% (Channel 5)
- Save the setting!

VCDS Release 10.6.1: 09-Cent. Elect., Adaptation

Daytime Running Lights Dimming (Low Beam)

92.0 %

Load

Channel

05

Up

Dn

Read

Add to Log

Stored value

62

New value

92

Up

Dn

Test value

92

Test

Save

Done, Go Back

Daytime Running Lights Dimming (Low Beam) X

Affective with Halogen only.

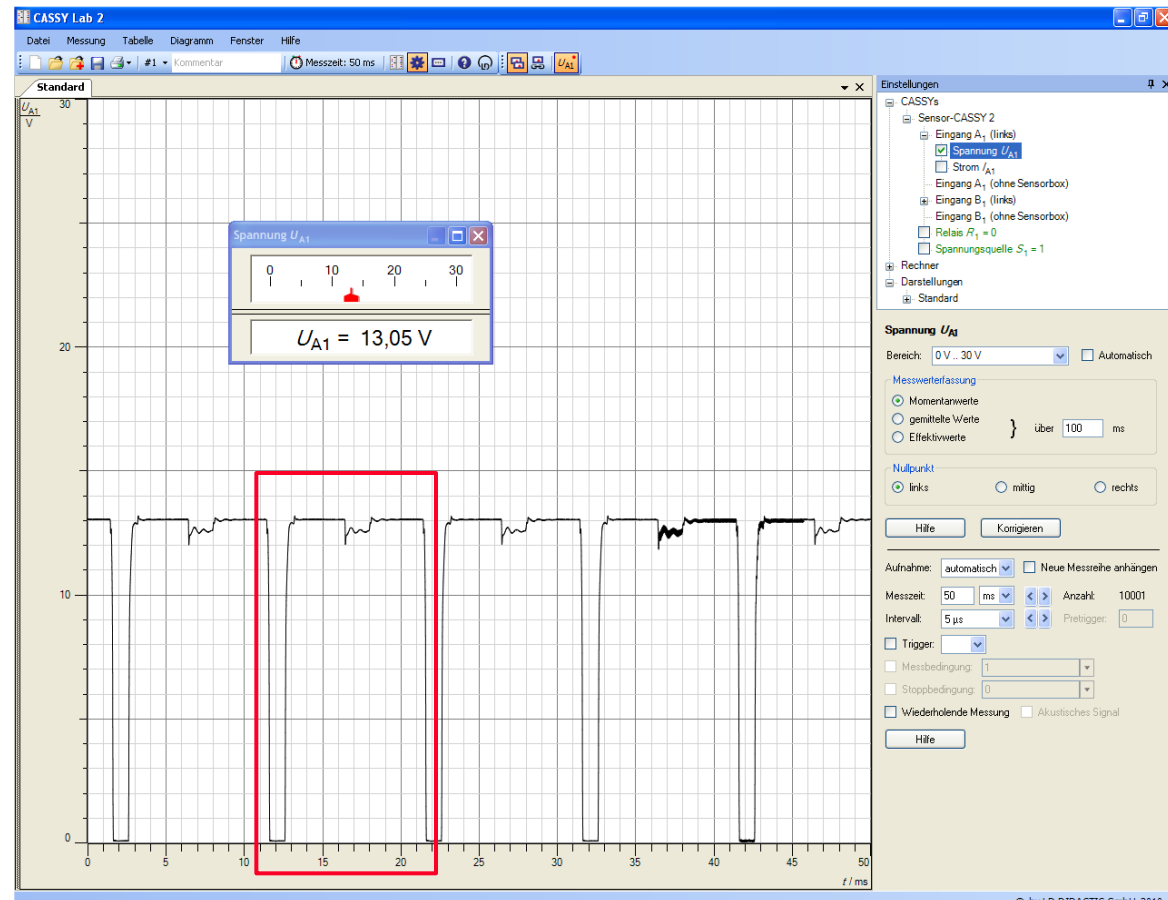
Standard: 92 %

Range: 51...100 %

DIAGNOSIS WITH VCDS

„Adaptation“

- Record the low beam signal left i. e. using Sensor CASSY 2
- Determine the frequency and the ratio of on-time and cycle time



DIAGNOSIS WITH VCDS

„Adaptation“

- Set the daytime running light dimming value to 62% (Channel 5)
- Save the setting!

VCDS Release 10.6.1: 09-Cent. Elect., Adaptation

Daytime Running Lights Dimming (Low Beam)

Tagfahrli	%	N/A	N/A
Text	Text		

Channel: 05 Up Dn Read Add to Log

Stored value: 62

new value: 62 Up Dn

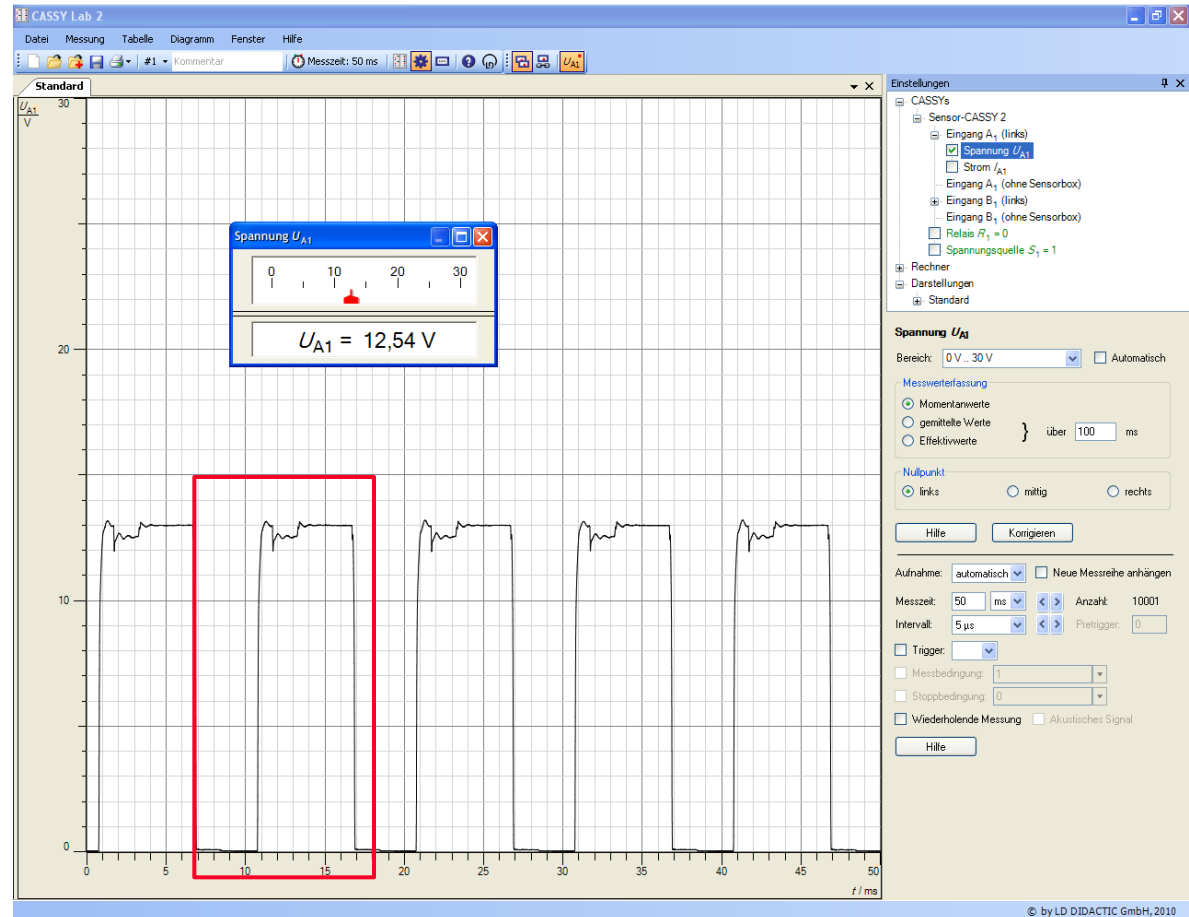
Test value:

Test Save Done, Go Back

DIAGNOSIS WITH VCDS

„Adaptation“

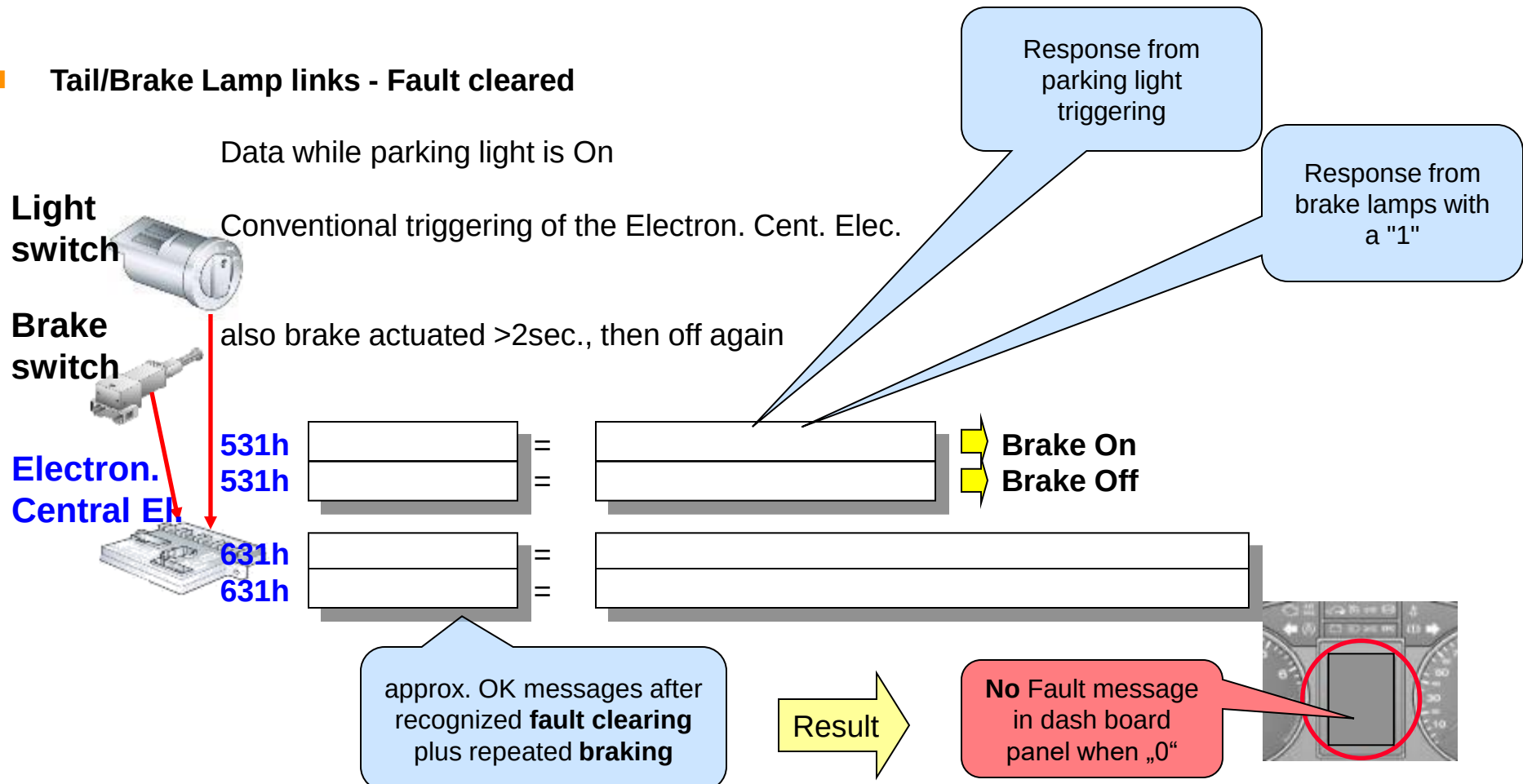
- Record the low beam signal left i. e. using Sensor CASSY 2
- Determine the frequency and the ratio of on-time and cycle time



Student's worksheets

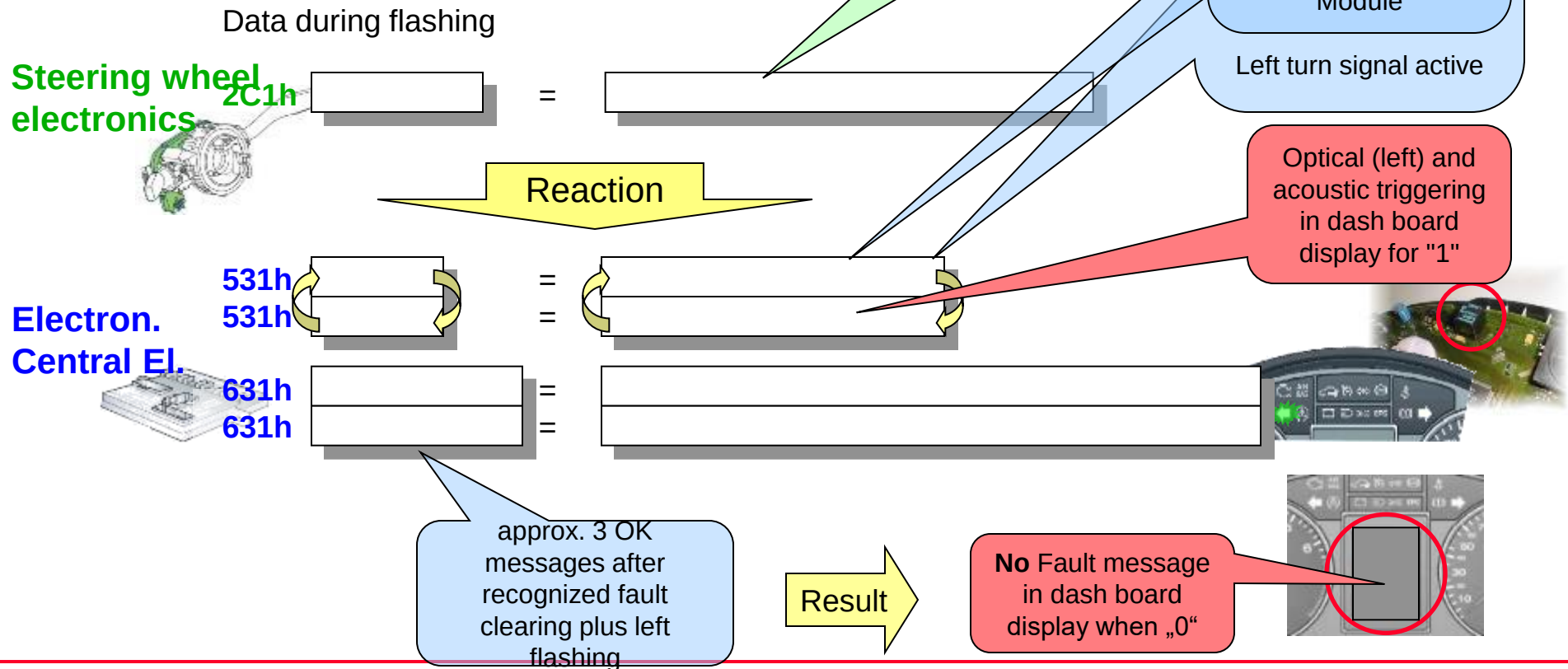
TAIL/BRAKE LIGHT

■ Tail/Brake Lamp links - Fault cleared



TURN SIGNAL CONTROL

■ Left front turn signal lamp - Fault cleared



HIGH BEAM CONTROL

■ Left high beam lamp - Fault cleared

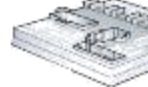
Steering wheel
electronics
2C1h



Data during High beam On

Electron.
Central El.

531h
631h
631h



Reaction

High beam On
by means of "1"

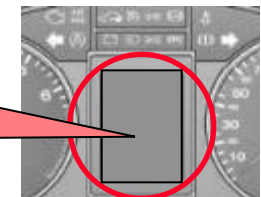
Response from
High beam triggering

Optical triggering
in dash board
display for "1"

High beam Off - On !!
approx. 3 OK messages
after recognized fault
clearing

Result

No Fault message
in dash board
display when „0“



LICENCE PLATE LAMP CONTROL

■ Licence plate illumination - Fault cleared

Data while parking light is On

Light switch

Conventional triggering of the Electron. Cent. Elec.

Electron. Central El.

531h

631h

631h

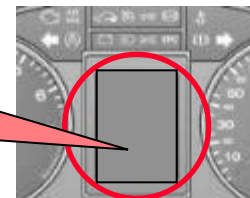
Response from parking light triggering

Optical triggering in dash board display for "1"

approx. 3 OK messages after recognized fault clearing

Result

No Fault message in dash board display when „0“



Equipment required

CAN-Bus

CAN-Bus to USB-Adapter,
available as **LEYBOLD** article 738581USB: CAN-Bus Adapter USB

PC-Measurement interface,
available as **LEYBOLD** article 524013SKFZ: Sensor-CASSY 2 Starter, Automotive

LIN-Bus

LIN-Bus to USB-Adapter,
available as **LEYBOLD** article 738588: LIN-Bus-PC-Adapter USB

PC-Measurement interface,
available as **LEYBOLD** article 524013SKFZ: Sensor-CASSY 2 Starter, Kfz

VCDS

Self diagnosis adapter,

available as **LEYBOLD** article 7379803: Kfz-Diagnose-Adapter CAN+USB

PC-Measurement interface,

available as **LEYBOLD** article 524013SKFZ: Sensor-CASSY 2 Starter, Kfz

Notes on coding

Coding of control units:

Central comfort system (46):	6720
Steering wheel electronics (16):	11
Central electronics (09):	11101

Adaptation of control units:

Dash panel insert (17):	1025
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Note: only with parts 8E0 920 933 P and 8E0 920 933 PX